

```
!pip install transformers pandas matplotlib seaborn
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.3)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.
Requirement already satisfied: shellingham in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->tr
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim->transformers) (0.
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->tra
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->tra
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-s
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from transformers import pipeline
```

```
emotion_classifier = pipeline(
    "text-classification",
    model="j-hartmann/emotion-english-distilroberta-base",
    return_all_scores=False
)

print("Emotion model loaded successfully 🟢")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), se
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(

config.json:      1.00k/? [00:00<00:00, 33.0kB/s]

pytorch_model.bin: 100%                               329M/329M [00:05<00:00, 114MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fas
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
model.safetensors: 61%                               201M/329M [00:07<00:08, 15.2MB/s]

Loading weights: 100%                               105/105 [00:00<00:00, 141.52it/s, Materializing param=roberta.encoder.layer.5.output.dense.weight]
RobertaForSequenceClassification LOAD REPORT from: j-hartmann/emotion-english-distilroberta-base
Key                | Status          | |
-----+-----+-----+
roberta.embeddings.position_ids | UNEXPECTED     | |

Notes:
- UNEXPECTED      :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%                           294/294 [00:00<00:00, 2.20kB/s]

vocab.json:      798k/? [00:00<00:00, 5.70MB/s]
merges.txt:      456k/? [00:00<00:00, 3.78MB/s]

tokenizer.json:   1.36M/? [00:00<00:00, 5.90MB/s]

special_tokens_map.json: 100%                         239/239 [00:00<00:00, 6.28kB/s]
Emotion model loaded successfully 

```

```

# Test the emotion model
test_sentence = "I am very disappointed with the teaching method."

result = emotion_classifier(test_sentence)

print("Text:", test_sentence)
print("Detected Emotion:", result[0]["label"])
print("Confidence Score:", round(result[0]["score"], 3))

```

```

Text: I am very disappointed with the teaching method.
Detected Emotion: sadness
Confidence Score: 0.855

```

```

# Sample student feedback data
feedback_data = [
    "The teacher explains concepts very clearly and I am happy.",
    "I feel confused during lectures.",
    "The course material is very helpful.",
    "I am frustrated because assignments are too difficult.",
    "The classroom environment is friendly and supportive.",
    "I am disappointed with the grading system.",
    "The lectures are boring sometimes.",
    "I really enjoy this subject."
]

df = pd.DataFrame(feedback_data, columns=["Feedback"])

df.head()

```

	Feedback
0	The teacher explains concepts very clearly and...
1	I feel confused during lectures.
2	The course material is very helpful.
3	I am frustrated because assignments are too di...
4	The classroom environment is friendly and supp...

```

# Function to detect emotion
def detect_emotion(text):
    result = emotion_classifier(text)

```

```

    return result[0]["label"]

# Apply emotion detection to all feedback
df["Emotion"] = df["Feedback"].apply(detect_emotion)

df

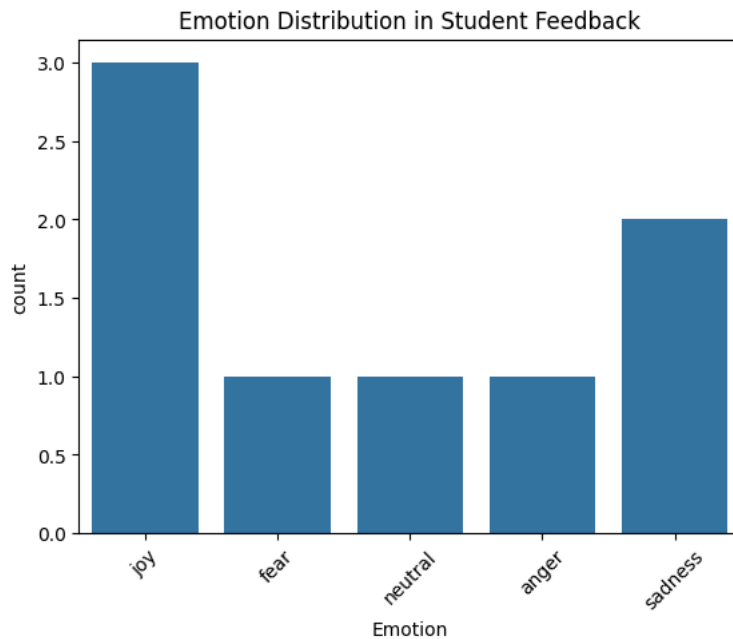
```

	Feedback	Emotion
0	The teacher explains concepts very clearly and...	joy
1	I feel confused during lectures.	fear
2	The course material is very helpful.	neutral
3	I am frustrated because assignments are too di...	anger
4	The classroom environment is friendly and supp...	joy
5	I am disappointed with the grading system.	sadness
6	The lectures are boring sometimes.	sadness
7	I really enjoy this subject.	joy

```

plt.figure()
sns.countplot(x="Emotion", data=df)
plt.title("Emotion Distribution in Student Feedback")
plt.xticks(rotation=45)
plt.show()

```



```

# Simple Intent Detection Function

def detect_intent(text):
    text_lower = text.lower()

    if any(word in text_lower for word in ["disappointed", "frustrated", "boring", "difficult", "confused"]):
        return "Complaint"

    elif any(word in text_lower for word in ["should", "improve", "recommend", "suggest"]):
        return "Suggestion"

    elif any(word in text_lower for word in ["happy", "enjoy", "helpful", "friendly"]):
        return "Appreciation"

    else:
        return "Neutral"

print("Intent function created ✅")

```

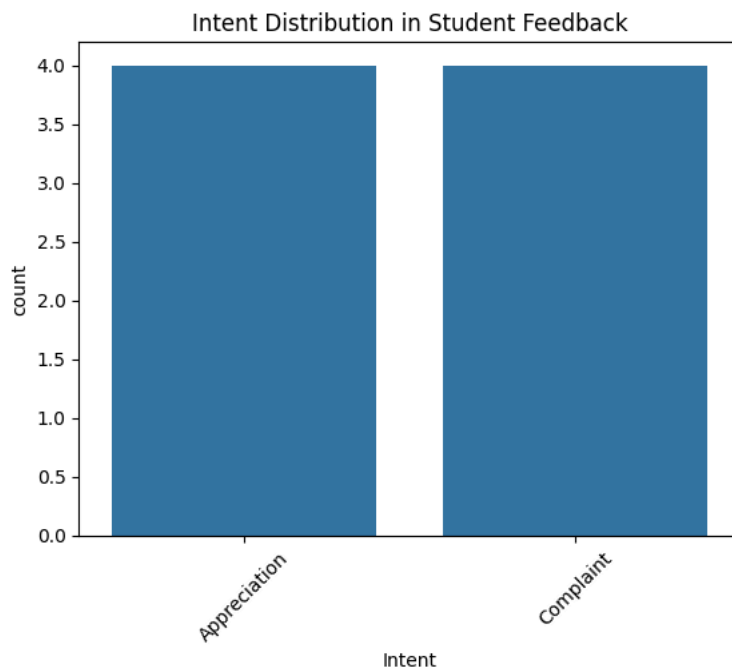
Intent function created ✅

```
# Apply intent detection
df["Intent"] = df["Feedback"].apply(detect_intent)

df
```

	Feedback	Emotion	Intent
0	The teacher explains concepts very clearly and...	joy	Appreciation
1	I feel confused during lectures.	fear	Complaint
2	The course material is very helpful.	neutral	Appreciation
3	I am frustrated because assignments are too di...	anger	Complaint
4	The classroom environment is friendly and supp...	joy	Appreciation
5	I am disappointed with the grading system.	sadness	Complaint
6	The lectures are boring sometimes.	sadness	Complaint
7	I really enjoy this subject.	joy	Appreciation

```
plt.figure()
sns.countplot(x="Intent", data=df)
plt.title("Intent Distribution in Student Feedback")
plt.xticks(rotation=45)
plt.show()
```



```
from transformers import pipeline

# Combine all feedback
all_feedback_text = " ".join(df["Feedback"])

# Load lightweight text generation model
generator = pipeline("text-generation", model="gpt2")

# Generate summary-style output
prompt = "Summarize the following student feedback in 5-6 lines:\n" + all_feedback_text

summary = generator(prompt, max_length=150, num_return_sequences=1)

print("AI Generated Summary:\n")
print(summary[0]["generated_text"])
```

```

config.json: 100%                               665/665 [00:00<00:00, 17.2kB/s]

model.safetensors: 100%                         548M/548M [00:06<00:00, 101MB/s]

Loading weights: 100%                           148/148 [00:00<00:00, 246.56it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key | Status | |
-----+-----+---
h.{0...11}.attn.bias | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

generation_config.json: 100%                    124/124 [00:00<00:00, 2.23kB/s]

tokenizer_config.json: 100%                     26.0/26.0 [00:00<00:00, 595B/s]

vocab.json: 100%                               1.04M/1.04M [00:00<00:00, 6.27MB/s]

merges.txt: 100%                              456k/456k [00:00<00:00, 2.79MB/s]

tokenizer.json: 100%                           1.36M/1.36M [00:00<00:00, 11.7MB/s]

Passing `generation_config` together with generation-related arguments(={'max_length', 'num_return_sequences'}) is deprecate
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length` (=150) seem to have been set. `max_new_tokens` will take precedence. Please re
AI Generated Summary:

```

Summarize the following student feedback in 5-6 lines:

The teacher explains concepts very clearly and I am happy. I feel confused during lectures. The course material is very hel
The teacher's advice is always helpful in the classroom. I am happy with the assignments and the course material.
The teacher's advice is always helpful in the classroom. I am happy with the assignments and the course material. I appreci
The teacher provides me with a personal support system. I am very happy with the assignments.
The teacher provides me with a personal support system. I am very happy with the assignments. I feel that the course materi
The teacher provides me with a personal support system. I am very happy with the assignments. I feel that the course materi
I feel that the course material is very nice. I am very satisfied with the material. I appreciate the feedback and I am not

```
!pip install streamlit pyngrok
```

```

Collecting streamlit
  Downloading streamlit-1.54.0-py3-none-any.whl.metadata (9.8 kB)
Collecting pyngrok
  Downloading pyngrok-7.5.0-py3-none-any.whl.metadata (8.1 kB)
Requirement already satisfied: altair!=5.4.0,!5.4.1,<7,>=4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (
Requirement already satisfied: blinker<2,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<7,>=5.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.2.6)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (8.3.1)
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in /usr/local/lib/python3.12/dist-packages (from streamlit) (3.
Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.0.2)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (26.0)
Requirement already satisfied: pandas<3,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.2.2)
Requirement already satisfied: pillow<13,>=7.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (11.3.0)
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Downloading pydeck-0.9.1-py2.py3-none-any.whl.metadata (4.1 kB)
Requirement already satisfied: protobuf<7,>=3.20 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.29.6)
Requirement already satisfied: pyarrow>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (18.1.0)
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: tenacity<10,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (9.1.4)
Requirement already satisfied: tomli<2,>=0.10.1 in /usr/local/lib/python3.12/dist-packages (from streamlit) (0.10.2)
Requirement already satisfied: tornado!=6.5.0,<7,>=6.0.3 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.5.1)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (4.
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.12/dist-packages (from pyngrok) (6.0.3)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=4.0->stre
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!5.4.1,<7,>
Requirement already satisfied: gitdb<5,>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from gitpython!=3.1.19,<4,>=3.0.
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<3,>=1.4.0->st
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<3,>=1.4.0->streamlit) (
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<3,>=1.4.0->streamlit)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27-
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->strea
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->strea
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from gitdb<5,>=4.0.1->gitpython!
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->altair!=5.4.0,!5.4.
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonsc
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<3,>
Downloading streamlit-1.54.0-py3-none-any.whl (9.1 MB)

```

Downloading pyngrok-7.5.0-py3-none-any.whl (24 kB)
 Downloading pydeck-0.9.1-py2.py3-none-any.whl (6.9 MB)
 6.9/6.9 MB 104.9 MB/s eta 0:00:00
 Installing collected packages: pyngrok, pydeck, streamlit
 Successfully installed pydeck-0.9.1 pyngrok-7.5.0 streamlit-1.54.0

```
%%writefile app.py

import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from transformers import pipeline

st.set_page_config(page_title="Emotion & Intent Analyzer", page_icon="📊")

st.title("📊 Student Feedback Emotion & Intent Analyzer")
st.write("AI-powered dashboard for analyzing student feedback.")

# Load emotion model
@st.cache_resource
def load_emotion_model():
    return pipeline(
        "text-classification",
        model="j-hartmann/emotion-english-distilroberta-base"
    )

emotion_classifier = load_emotion_model()

# Emotion detection
def detect_emotion(text):
    result = emotion_classifier(text)
    return result[0]["label"]

# Intent detection
def detect_intent(text):
    text_lower = text.lower()

    if any(word in text_lower for word in ["disappointed", "frustrated", "boring", "difficult", "confused"]):
        return "Complaint"
    elif any(word in text_lower for word in ["should", "improve", "recommend", "suggest"]):
        return "Suggestion"
    elif any(word in text_lower for word in ["happy", "enjoy", "helpful", "friendly"]):
        return "Appreciation"
    else:
        return "Neutral"

# User input
st.subheader("Enter Student Feedback (one per line):")
user_input = st.text_area("Paste feedback here...")

if st.button("Analyze"):
    if user_input:
        feedback_list = user_input.split("\n")
        df = pd.DataFrame(feedback_list, columns=["Feedback"])

        df["Emotion"] = df["Feedback"].apply(detect_emotion)
        df["Intent"] = df["Feedback"].apply(detect_intent)

        st.subheader("📊 Analysis Results")
        st.dataframe(df)

        # Emotion chart
        st.subheader("📊 Emotion Distribution")
        fig1 = plt.figure()
        sns.countplot(x="Emotion", data=df)
        plt.xticks(rotation=45)
        st.pyplot(fig1)

        # Intent chart
        st.subheader("📊 Intent Distribution")
        fig2 = plt.figure()
        sns.countplot(x="Intent", data=df)
        st.pyplot(fig2)
```

Writing app.py

```
from pyngrok import ngrok

ngrok.set_auth_token("3A9sTtNVpy86Tj371iYLz1VaEc3_5bKVxFNs7Nd4b3o5BvG9e")
```

```
from pyngrok import ngrok
import os

!kill streamlit
get_ipython().system_raw('streamlit run app.py &')

public_url = ngrok.connect(8501)
print("Your Dashboard is Live at:")
print(public_url)
```

Your Dashboard is Live at:
 NgrokTunnel: "<https://squintingly-unbridged-lylah.ngrok-free.dev>" -> "<http://localhost:8501>"

```
#More Professional
%%writefile app.py

import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from transformers import pipeline

st.set_page_config(page_title="Emotion & Intent Analyzer", page_icon="🎨", layout="wide")

st.title("🎨 AI-Powered Student Feedback Analyzer")
st.markdown("### Emotion Detection • Intent Classification • AI Summary")

# Load models
@st.cache_resource
def load_emotion_model():
    return pipeline(
        "text-classification",
        model="j-hartmann/emotion-english-distilroberta-base"
    )

@st.cache_resource
def load_generator():
    return pipeline("text-generation", model="gpt2")

emotion_classifier = load_emotion_model()
generator = load_generator()

# Emotion function
def detect_emotion(text):
    result = emotion_classifier(text)
    return result[0]["label"]

# Intent function
def detect_intent(text):
    text_lower = text.lower()

    if any(word in text_lower for word in ["disappointed", "frustrated", "boring", "difficult", "confused"]):
        return "Complaint"
    elif any(word in text_lower for word in ["should", "improve", "recommend", "suggest"]):
        return "Suggestion"
    elif any(word in text_lower for word in ["happy", "enjoy", "helpful", "friendly"]):
        return "Appreciation"
    else:
        return "Neutral"

# Input section
st.subheader("Enter Student Feedback (one per line)")
user_input = st.text_area("Paste feedback here...")

if st.button("Analyze Feedback"):
    if user_input:
        feedback_list = user_input.split("\n")
        df = pd.DataFrame(feedback_list, columns=["Feedback"])
```

```

df["Emotion"] = df["Feedback"].apply(detect_emotion)
df["Intent"] = df["Feedback"].apply(detect_intent)

st.subheader("📄 Analysis Table")
st.dataframe(df)

col1, col2 = st.columns(2)

with col1:
    st.subheader("📊 Emotion Distribution")
    fig1 = plt.figure()
    sns.countplot(x="Emotion", data=df)
    plt.xticks(rotation=45)
    st.pyplot(fig1)

with col2:
    st.subheader("📊 Intent Distribution")
    fig2 = plt.figure()
    sns.countplot(x="Intent", data=df)
    st.pyplot(fig2)

# AI Summary
st.subheader("🤖 AI Generated Summary")

combined_text = " ".join(df["Feedback"])
prompt = "Summarize the following student feedback in 5 lines:\n" + combined_text

summary = generator(prompt, max_length=150, num_return_sequences=1)
st.write(summary[0]["generated_text"])

```

Overwriting app.py

```

!kill streamlit
get_ipython().system_raw('streamlit run app.py &')

```

```

from pyngrok import ngrok
public_url = ngrok.connect(8501)
print(public_url)

```